

Exposé I. Presheaves

A. Grothendieck and J.-L. Verdier

In numbers 0 through 5 of this exposé we present the elementary, and most often well-known, properties of categories of presheaves.¹ These properties are used constantly throughout the rest of the seminar, and knowing them is essential for understanding the subsequent exposés. The proofs are immediate; for the most part they are omitted. In numbers 6 through 9 we take up a few themes that are used several times later on. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki.

1 Universes

A *universe* is a nonempty set² $\mathcal{U}$ satisfying the following properties:

- (U1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.
- (U2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.
- (U3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.
- (U4) If $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\bigcup_{i \in I} x_i \in \mathcal{U}$.

From the preceding axioms one easily deduces the following properties:

- if $x \in \mathcal{U}$, then the singleton $\{x\}$ belongs to $\mathcal{U}$;
- if x is a subset of some $y \in \mathcal{U}$, then $x \in \mathcal{U}$;
- if $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$, in Kuratowski's definition, is an element of $\mathcal{U}$;
- if $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$;
- if $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ indexed by an element $I \in \mathcal{U}$, then the product $\prod_{i \in I} x_i$ is an element of $\mathcal{U}$;
- if $x \in \mathcal{U}$, then $\text{card}(x) < \text{card}(\mathcal{U})$ strictly. In particular, the relation $\mathcal{U} \in \mathcal{U}$ is not satisfied.

Thus one may perform all the usual operations of set theory starting from elements of a universe without the final result ceasing to be an element of that universe.

The first use of the notion of universe is that it supplies a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}\text{-Set}$), the category of topological spaces belonging to $\mathcal{U}$, the category of commutative groups belonging to $\mathcal{U}$ ($\mathcal{U}\text{-Ab}$), the category of categories belonging to $\mathcal{U}$, and so on.

¹The reader may also consult SGA 3, I, §§1–3.

²This definition contradicts Bourbaki's appendix below. However, the interest of empty universes seems debatable.

However, the only known universe is the set of symbols of the type $\{\{\emptyset\}, \{\{\emptyset\}, \emptyset\}\}$, etc. All the elements of this universe are finite sets, and this universe is countable. In particular, no universe is known that contains an element of infinite cardinality. We are therefore led to add to the axioms of set theory the axiom

($\mathcal{U}A$) *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom ($\mathcal{U}A$) is independent of the axioms of set theory.

We shall also add the axiom:

($\mathcal{U}B$) *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then*

The consistency of axioms ($\mathcal{U}A$) and ($\mathcal{U}B$) relative to the other axioms of set theory is not proved and, it seems, not provable.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the least upper bound of the cardinals of elements of $\mathcal{U}$; thus $c(\mathcal{U}) \leq \text{card}(\mathcal{U})$. The cardinal $c(\mathcal{U})$ has the following properties:

(**FI**) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(**FII**) If $(a_i)_{i \in I}$ is a family of cardinals strictly smaller than $c(\mathcal{U})$, and if $\text{card}(I)$ is strictly smaller than $c(\mathcal{U})$, then $\sum_{i \in I} a_i < c(\mathcal{U})$.

Cardinals satisfying properties (FI) and (FII) are called *strongly inaccessible cardinals*.

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom ($\mathcal{U}A$) implies:

($\mathcal{U}A'$) *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

One can show (11) that, conversely, the consistency of ($\mathcal{U}A'$) implies the consistency of ($\mathcal{U}A$), and that the consistency of these axioms implies the consistency of axiom ($\mathcal{U}B$).

Call a set E *Artinian* if there is no infinite family $(x_n)_{n \in \mathbb{N}}$ such that $x_0 \in E$ and $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between strongly inaccessible cardinals and Artinian universes, defined as follows: to every strongly inaccessible cardinal c one associates the unique Artinian universe $\mathcal{U}_c$ such that

$$\text{card}(\mathcal{U}_c) = c.$$

If $c < c'$ are two strongly inaccessible cardinals, then

$$\mathcal{U}_c \in \mathcal{U}_{c'}.$$

In particular, the Artinian universes of cardinal smaller than a given cardinal form a well-ordered set for the membership relation. Axiom ($\mathcal{U}A'$) is equivalent to the axiom:

($\mathcal{U}A''$) *Every Artinian set is an element of an Artinian universe.*

We note that all the usual sets $(\mathbb{Z}, \mathbb{Q}, \dots)$ are Artinian sets.

2 $\mathcal{U}$ -categories. Presheaves of sets

1.0

In the rest of the seminar, unless the contrary is expressly stated, the universes considered will possess an element of infinite cardinality. Let $\mathcal{U}$ be a universe. A set is said to be $\mathcal{U}$ -small, or simply *small* when no confusion can result, if it is *isomorphic* to an element of $\mathcal{U}$. We also use the terminology small group, small ring, small category,³ etc. We shall often suppose, without explicit mention, that the schemes, topological spaces, indexing sets, etc. with which we work are in $\mathcal{U}$, or at least have cardinality in $\mathcal{U}$; however, many of the categories with which we shall work will not themselves be in $\mathcal{U}$.

Definition (1.1). Let $\mathcal{U}$ be a universe and let C be a category. We say that C is a $\mathcal{U}$ -category if, for every pair (x, y) of objects of C , the set $\text{Hom}_C(x, y)$ is $\mathcal{U}$ -small.

1.1.1

Let C and D be two categories, and let $\text{Fun}(C, D)$ denote the category of functors from C to D . The following assertions are immediate:

- (a) If C and D are elements of a universe $\mathcal{U}$ (respectively are $\mathcal{U}$ -small), then the category $\text{Fun}(C, D)$ is an element of $\mathcal{U}$ (respectively is $\mathcal{U}$ -small).
- (b) If C is $\mathcal{U}$ -small and D is a $\mathcal{U}$ -category, then $\text{Fun}(C, D)$ is a $\mathcal{U}$ -category.

Remark (1.1.2). Let D be a category satisfying the following properties:

- (C1) the set $\text{Ob}(D)$ is contained in the universe $\mathcal{U}$;
- (C2) for every pair (x, y) of objects of D , the set $\text{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}\text{-Set}$, $\mathcal{U}\text{-Ab}$, and so on. Let C be a category belonging to $\mathcal{U}$. Then the category $\text{Fun}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\text{Fun}(C, \mathcal{U}\text{-Set})$ has neither property (C1) nor property (C2). This justifies the definition adopted for a $\mathcal{U}$ -category, rather than the more restrictive notion given by conditions (C1) and (C2) above.

Definition (1.2). Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$ —or, when no confusion can result, the category of presheaves on C —is the category of contravariant functors on C with values in the category of $\mathcal{U}$ -sets.

We denote by $\widehat{C}_{\mathcal{U}}$, or more simply by $\widehat{C}$ when no confusion can result, the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$ -presheaves, or simply presheaves, on C . When C is $\mathcal{U}$ -small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$ -category. When C is a $\mathcal{U}$ -category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$ -category.

Construction–Definition (1.3). Let x be an object of a $\mathcal{U}$ -category C . The $\mathcal{U}$ -functor represented by x is the functor

$$h_{\mathcal{U}}(x) : C^{\text{op}} \longrightarrow \mathcal{U}\text{-Set}$$

constructed as follows.⁴ Let y be an object of C .

³A category C is viewed as a set of arrows, equipped with the subset of objects, viewed as the set of identity arrows, and with the source and target maps. Thus the expressions ‘ C is an element of $\mathcal{U}$ ’ and ‘ C is $\mathcal{U}$ -small’ make sense.

⁴ C^{op} will always denote the category opposite to C .

(a) If $\text{Hom}_C(y, x)$ is an element of $\mathcal{U}$, then

$$h_{\mathcal{U}}(x)(y) = \text{Hom}_C(y, x).$$

(b) Suppose that $\text{Hom}_C(y, x)$ is not an element of $\mathcal{U}$, and let $R(Z, x, y)$ be the relation: ‘The set Z is the target of an isomorphism $\text{Hom}_C(y, x) \xrightarrow{\sim} Z$.’ We then put

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

By axiom $(\mathcal{U}B)$, $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$. Let $R'(u, x, y)$ be the relation: ‘ u is a bijection

$$u : \text{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y)'.$$

We put

$$\varphi(y, x) = \tau_u R'(u).$$

Notice that in both cases (a) and (b) we have a canonical isomorphism

$$\varphi(y, x) : \text{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

In case (a), $\varphi(y, x)$ is the identity. Now let $u : y \rightarrow y'$ be an arrow of C . Composition of morphisms defines a map

$$\text{Hom}_C(u, x) : \text{Hom}_C(y', x) \longrightarrow \text{Hom}_C(y, x).$$

We set

$$h_{\mathcal{U}}(x)(u) = \varphi(y, x) \text{Hom}_C(u, x) \varphi(y', x)^{-1}.$$

One verifies immediately that $h_{\mathcal{U}}(x)$, so defined, is a functor $C^{\text{op}} \rightarrow \mathcal{U}\text{-Set}$.

1.3.1

Similarly, using the isomorphisms φ , for every morphism $v : x \rightarrow x'$ one defines a morphism of functors

$$h_{\mathcal{U}}(v) : h_{\mathcal{U}}(x) \longrightarrow h_{\mathcal{U}}(x'),$$

and one verifies immediately that this defines a functor

$$h_{\mathcal{U}} : C \longrightarrow \hat{C}_{\mathcal{U}}.$$

1.3.2

Let $\mathcal{V}$ now be a universe such that $\mathcal{U} \subset \mathcal{V}$. We then have a canonical fully faithful functor

$$\mathcal{U}\text{-Set} \hookrightarrow \mathcal{V}\text{-Set},$$

and hence a fully faithful functor

$$\hat{C}_{\mathcal{U}} \hookrightarrow \hat{C}_{\mathcal{V}}.$$

The diagram

$$\begin{array}{ccc} C & \xrightarrow{h_{\mathcal{U}}} & \hat{C}_{\mathcal{U}} \\ & \searrow h_{\mathcal{V}} & \downarrow \\ & & \hat{C}_{\mathcal{V}} \end{array}$$

commutes up to canonical isomorphism. When C is an element of $\mathcal{U}$, this canonical isomorphism is the identity.

1.3.3

In practice, the universe $\mathcal{U}$ is fixed once and for all and is not mentioned. We then use the notation $\widehat{C}$ for the category of $\mathcal{U}$ -presheaves of sets and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C , the presheaf $h(x)$ is called the *presheaf represented by x* , and we shall always identify its value at an object $y \in \text{Ob}(C)$ with $\text{Hom}_C(y, x)$.

Proposition (1.4). Let C be a $\mathcal{U}$ -category, let F be a presheaf on C , and let X be an object of C . There is an isomorphism, functorial in X and in F ,

$$i : \text{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where, when F is of the form $h(Y)$, the map i is just the map

$$h : \text{Hom}(h(X), h(Y)) \longleftarrow \text{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1

This proposition justifies the usual abuse of language that identifies an object of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h , or, using the above abuse of language, isomorphic to an object of C , is called a *representable presheaf*.

1.4.2

Let $\mathcal{V}$ be a universe containing a universe $\mathcal{U}$. The category of $\mathcal{U}$ -presheaves is a *full* subcategory of the category of $\mathcal{V}$ -presheaves; consequently, a $\mathcal{U}$ -presheaf is representable if and only if its image in the category of $\mathcal{V}$ -presheaves is a representable $\mathcal{V}$ -presheaf.